VEXON DEVELOPER HANDBOOK · GUIDE 10

VEXON Developer Handbook - Master Reference

Open this first. The short map of the whole system, the commands, the checklists, and where every detailed guide lives

Written for	Anyone who needs to understand, run, debug or maintain VEXON quickly
Project	VEXON / CampusConnect (repository: this folder's parent project)
Evidence	Generated from a direct inspection of the repository source code
Repository state	branch fix/csrf-legacy-cookie, commit af623fb, plus 126 uncommitted working-tree change(s)
Generated	16 October 2026, 21:49
Rebuild	python docs/developer-handbook/build_handbook.py

Accuracy rule: every statement about VEXON in this guide is based on files in the repository. Anything that could not be confirmed is labelled 'unverified', 'planned' or 'hypothetical'. No secrets, passwords or real student data are reproduced.

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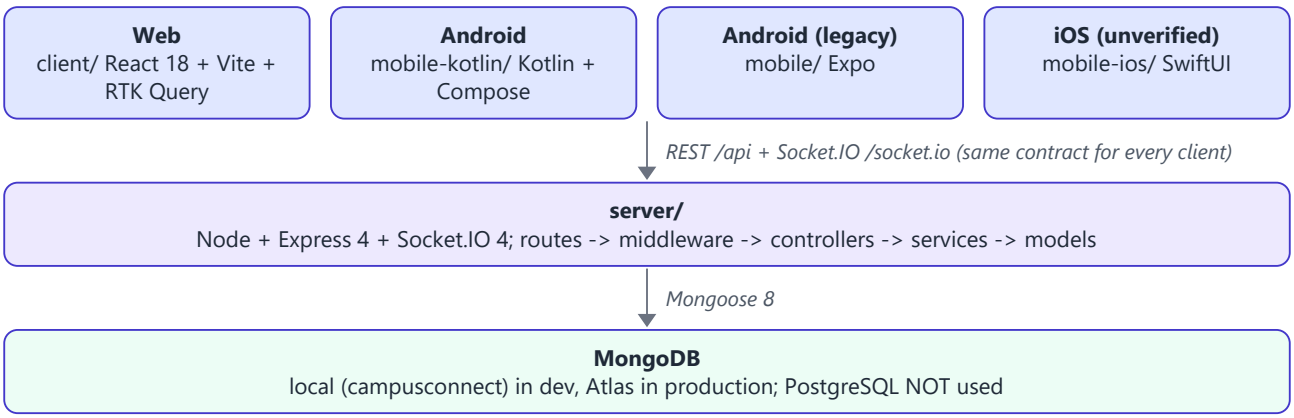
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How to use the handbook

#	Guide (PDF in `docs/developer-handbook/`)	Open it when you need to...
10	`10_VEXON_Developer_Handbook.pdf` (this)	Find anything fast
01	`01_VEXON_Project_Overview.pdf`	Understand the architecture, security model and apps
02	`02_JavaScript_TypeScript_React_Guide.pdf`	Read or change the web dashboard
03	`03_Node_Express_Backend_Guide.pdf`	Read, debug or extend the API
04	`04_Database_MongoDB_PostgreSQL_Guide.pdf`	Work with data, backups, or plan PostgreSQL
05	`05_VEXON_Feature_By_Feature_Guide.pdf`	Debug one module (files, routes, models, tests)
06	`06_Debugging_and_Error_Recovery_Guide.pdf`	Something is broken
07	`07_Testing_Git_and_Safe_Code_Changes.pdf`	Change code safely, run tests, commit
08	`08_Deployment_and_Production_Maintenance.pdf`	Deploy, configure production, plan the VPS
09	`09_VEXON_30_Day_Learning_Workbook.pdf`	Learn the codebase step by step
11	`11_Error_Messages_Troubleshooting_Guide.pdf`	A user reports an error message: what it means and what to do (every server message is listed)

Every PDF is generated from an editable Markdown file in `docs/developer-handbook/src/`. Rebuild them all with `python docs/developer-handbook/build_handbook.py`. Validate them with `python docs/developer-handbook/validate_handbook.py`. `INDEX.md` and `PROJECT_STATUS.md` sit next to the PDFs.

Architecture summary



- **The server is the only authority.** Roles (15, in `server/src/constants.js`), scope and validation are enforced there. Client guards are cosmetic.
- **Auth:** a 15-minute access JWT bound to a `Session` row, plus a rotating 7-day refresh token (an httpOnly cookie on web, the JSON body on mobile). CSRF double-submit for browsers. A 2-minute idle rule for non-exempt web roles.
- **Real-time:** sockets announce "something changed", and clients re-read over REST. Chat and bus positions are the exceptions.

Folder navigation map

I am looking for...	Go to
Server start-up and timers	`server/src/server.js`
Middleware order, CSP, CORS, static hosting	`server/src/app.js`
Environment variables and defaults	`server/src/config/env.js`, `server/.env.example`
Every API route and its validation	`server/src/routes/index.js`, `server/src/routes/alumniRoutes.js`
Who is allowed (role lists)	`server/src/constants.js` (server), `client/src/utils/constants.js` (UI mirror)
Sign-in, sessions, tokens	`server/src/controllers/authController.js`, `server/src/utils/tokens.js`, `server/src/middleware/auth.js`
CSRF and origin checks	`server/src/middleware/security.js`
Error responses	`server/src/middleware/error.js`, `server/src/utils/http.js`
Notifications and push	`server/src/utils/notify.js`, `server/src/services/pushService.js`
Socket rooms and events	`server/src/config/socket.js`, `client/src/services/socket.js`, `client/src/components/layout/AppLayout.jsx`
Data models	`server/src/models/` (59 files)
Web routes and guards	`client/src/App.jsx`
Web API calls (all of them)	`client/src/services/api.js`
A web page	`client/src/pages/<module>/`
Android networking and session	`mobile-kotlin/app/src/main/java/com/vexon/campus/core/Http.kt`, `SecureStore.kt`
Android screens and routes	`mobile-kotlin/app/src/main/java/com/vexon/campus/screens/` (`Routes.kt`)
Deployment	`render.yaml`, `.github/workflows/`
Tests	`server/tests/`, `client/tests/`, `mobile/tests/`, `mobile-kotlin/app/src/test/`

Common development commands

```bash
npm run install:all                      # install root + server + client
npm run dev                              # API :5000 + web :5173 (needs MongoDB running)
npm run seed                             # DEV ONLY - wipes and reloads campusconnect
npm run seed:academic                    # adds demo academic data without wiping
npm test --prefix server                 # server integration tests (uses cmp_test)
npm test --prefix client                 # client unit tests
npm run build                            # build client/dist
npm run verify:bundle                    # scan client/dist for secrets / source maps
npm start                                # production-style start (serves client/dist + API)
node server/scripts/realtime-smoke.mjs   # live web<->mobile chat check (server running)
node server/scripts/simulate-bus.mjs --device SIM-TEST-1 --route "<route>"   # synthetic GPS, dev only
npm run migrate:break-periods --prefix server            # dry run (add -- --apply to write)
cd mobile-kotlin && gradlew.bat assembleDebug            # Android debug APK
cd mobile-kotlin && gradlew.bat testDebugUnitTest        # Android unit tests
python docs/developer-handbook/build_handbook.py        # rebuild these PDFs
```

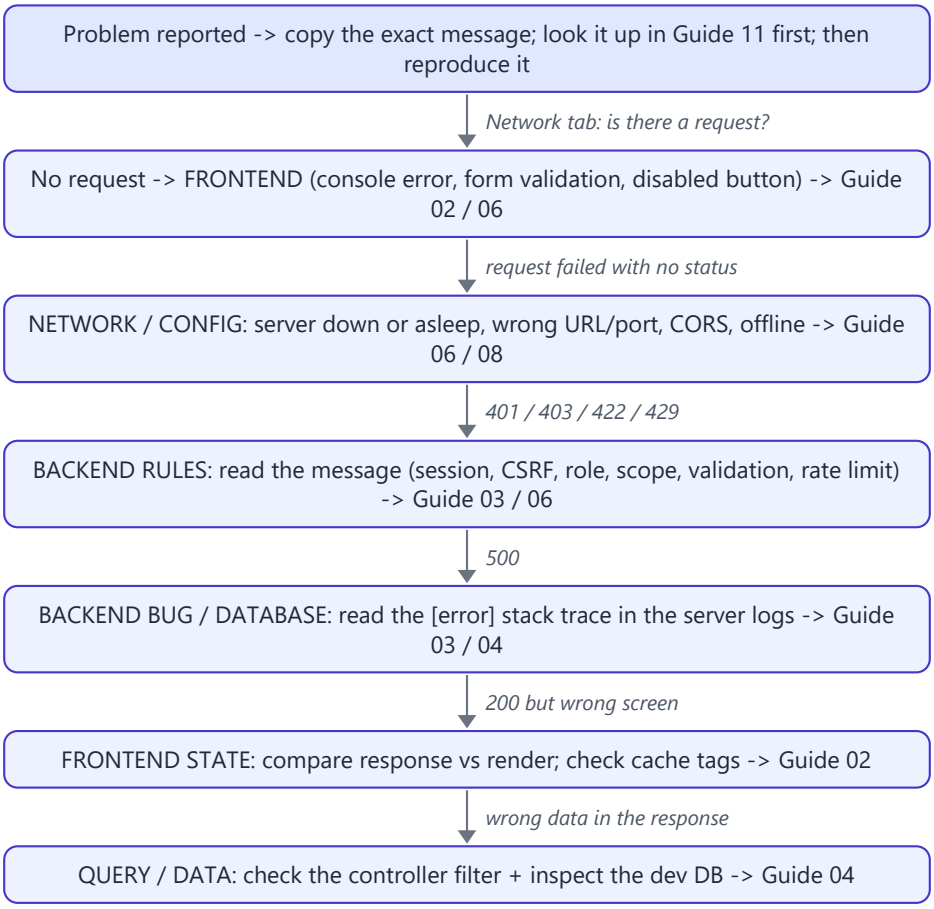
Critical configuration locations

Setting	Where it is set	Notes
Database choice	`DB_TARGET`, `MONGO_URI`, `MONGO_URI_ATLAS` in `server/.env` / the host	Local is the default
Secrets	`JWT_ACCESS_SECRET`, `JWT_REFRESH_SECRET`, `CHAT_ENCRYPTION_KEY`	Required in production; generated by Render
Public URL / origins	`CLIENT_URL`, `APP_URL`	Must match the real URL (CSRF/origin guard, Socket.IO)
Proxy hops	`TRUST_PROXY`	1 on Render
SMS / OTP	`OTP_SMS_ENABLED`, `SMS_*`	SMS is ON in `render.yaml`, over plain HTTP by explicit opt-in
Map	`MAP_STYLE_URL`, `MAP_CSP_ORIGINS`	Default OpenFreeMap
Android backend URL	`mobile-kotlin/gradle.properties` `vexon.apiUrl`	Currently the deployed URL, for **both** debug and release
Dev proxy	`client/vite.config.js` (`VITE_API_PROXY`)	Defaults to `http://localhost:5000`

Important API and database entry points

Need	API	Controller	Model(s)
Sign in / refresh	`POST /api/auth/login`, `/auth/refresh`	`authController.js`	`User`, `Session`
Current user	`GET /api/auth/me`	`authController.js`	`User`
Dashboard	`GET /api/dashboard`	`dashboardController.js`	many
Mark attendance	`POST /api/attendance/mark`	`attendanceController.js`	`AttendanceSession`, `AttendanceRecord`
Timetable	`GET /api/timetable`	`timetableController.js`	`TimetableSlot`, `Subject`
Gate pass	`POST /api/gate-pass`, `PATCH /api/gate-pass/:id/*-review`	`gatePassController.js`	`GatePass`, `OutpassOtp`
Complaint	`POST /api/complaints`	`complaintController.js`	`Complaint`, `ComplaintHistory`
Chat	`/api/chat/conversations/:id/messages`	`chatController.js`	`Conversation`, `Message`
Bus map	`GET /api/transport/buses`, socket `transport:subscribe`	`transportController.js`	`Bus`, `BusRoute`, `BusLocation`
SOS	`POST /api/sos`	`sosController.js`	`SOSAlert`
Health	`GET /api/health`	`app.js`	none

Troubleshooting decision tree



Testing checklist

- [ ] `npm test --prefix server`: all pass (MongoDB running).
- [ ] `npm test --prefix client`: all pass.
- [ ] `npm run build --prefix client` succeeds; `npm run verify:bundle` passes.
- [ ] Kotlin `testDebugUnitTest` passes (if Android changed).
- [ ] A new regression test exists for every bug fixed.
- [ ] Role checks re-tested for any route whose `authorize(...)` or scope changed.
- [ ] Manual smoke test of the affected workflow, as the affected roles.

Git safety checklist

- [ ] `git status` before starting. Commit or stash the existing work first (more than 120 uncommitted changes existed at audit time).
- [ ] One branch per change (`git switch -c fix/...`).
- [ ] Read the whole `git diff` before staging; stage named files, not `git add .`.
- [ ] No `.env`, keys, dumps, generated personal-data PDFs or `local.properties` staged.
- [ ] Commit messages explain *why*.
- [ ] Never `git reset --hard` or `git push --force` without a backup or stash and a clear reason.

Deployment and recovery checklist

- [ ] CI green on the exact commit.
- [ ] Production variables set (names in Guide 08); `CLIENT_URL` and `TRUST_PROXY` correct.
- [ ] Fresh database backup **and** a recently tested restore.
- [ ] Deploy, then check `/api/health`, sign-in, and one critical workflow per module touched.
- [ ] Watch the logs for `[error]` for the first hour.
- [ ] Rollback ready: the previous good commit id; how to redeploy it; data rollback only from backup.

Known issues and unverified areas

Item	Status	Detail
Server test suite	718 of 721 passed (16 Oct 2026)	2 date-dependent failures in `gate-exit-campus-day.test.js` (mixed clocks) and 1 intermittent file failure; details in Guide 07
PostgreSQL	Planned, not implemented	Only MongoDB exists (Guide 04 has a migration plan)
TypeScript	Not used	All code is JavaScript or JSX
iOS app	Unable to verify	Never compiled (per its README); untracked in git; needs a Mac
Browser E2E tests	Missing	No Playwright or Cypress; UI flows are tested manually
Clubs, discussions	Untested	No direct API tests
Real SMS, real push, real GPS trackers	Not verified in the audit	Code paths exist; tests use test providers or synthetic data
`/api/health`	Limited	Does not check MongoDB
Error monitoring	Missing	No error-tracking service configured
Backups	Manual only	No backup automation in the repository
SMS over HTTP	Accepted risk	`SMS_ALLOW_INSECURE_HTTP=true` in `render.yaml`
Android release signing	Debug key	Not suitable for a store release
Kotlin debug build URL	Points to production by default	Override with `-Pvexon.apiUrl` for local work
README drift	Several mismatches	Lockout rule, idle exemptions, service name, Android client, uploads (Guide 01)
Large web chunk	Build warning	`BusMap` chunk is about 1.05 MB minified
Uncommitted work	Risk	More than 120 changed files and the untracked `mobile-ios/` at audit time

Glossary (short)

API: URLs a program calls. **Middleware**: a function run before a route handler. **Controller**: the route's main logic. **Model**: a Mongoose schema for one collection. **JWT**: a signed token. **Session**: a per-device sign-in record. **CSRF**: a cross-site forged request. **RBAC**: role-based access control. **Scope check**: "is this record yours / your department's?". **RTK Query**: the web app's data layer. **Cache tag**: a label that triggers a refetch. **Socket room**: a group of connections receiving the same events. **TTL index**: auto-deletes old documents. **CI**: automated checks on GitHub. **Seed**: demo data loader (it wipes the database).

Source references

`README.md` · `render.yaml` · `server/src/server.js` · `server/src/app.js` · `server/src/config/env.js` · `server/src/routes/index.js` · `server/src/constants.js` · `server/src/middleware/` · `server/src/models/` · `client/src/App.jsx` · `client/src/services/api.js` · `mobile-kotlin/gradle.properties` · `mobile-kotlin/README.md` · `mobile-ios/README.md` · `.github/workflows/ci.yml` · `docs/developer-handbook/src/` · `docs/developer-handbook/build_handbook.py`